Appendix 3 exercises

1. Give an explicit bijection proving integers countable. State whether natural numbers start at zero or one.
2. For $\pi(m, n) = (m + n)(m + n + 1)/2 + m, m, n \in \mathbb{N}_0$, list the first two diagonals and explain why the construction enumerates all pairs.
3. For countably infinite S and $A \subseteq S$, prove A countable, separating finite and infinite cases.
4. Determine when unions of two, finitely many, and countably infinitely many countable sets are countable. In the infinite case, distinguish simultaneous specified enumerations from mere knowledge of each set's countability.
5. Prove $\mathcal{P}(\mathbb{N})$ uncountable by assuming a complete listing, defining its diagonal set, and deriving a contradiction.
6. Prove $(0, 1]$ uncountable, resolving eventually-all-nine decimal nonuniqueness before diagonalization.
7. Determine countability of rationals, irrationals, finite subsets of natural numbers, and all subsets of natural numbers. Prove each claim.
8. Give the general inverse of pairing and invert indices 18 and 35. Prove each index has exactly one preimage rather than only listing a finite table.
9. Bijection finite subsets of $\mathbb{N}_0$ with $\mathbb{N}_0$ through binary support. Encode $\{0, 2, 5\}$, invert 45, and explain why infinite subsets cannot all be encoded as finite integers this way.
10. Prove Cantor's theorem for $f : A \to \mathcal{P}(A)$, including empty A. Explain how singleton injection and impossibility of surjection establish a strict cardinal inequality.
11. Verify all branches, cross-branch injectivity, and surjectivity of the lecture's CSB construction. For $A = \mathbb{N}_0, B = 2\mathbb{N}_0$, choose suitable injections f, g.
12. Prove the ternary zero/two encoding injective by bounding the entire tail after the first differing digit. Construct the reverse injection using rational cuts and conclude equal cardinalities.
13. Prove algebraic reals countable and transcendental reals uncountable. Explain how countably many finite programs and uncountably many binary functions imply noncomputable functions exist, without deciding computability of any specified function.
14. Compute $\phi(0), \phi(2), \phi(-2)$ and check the inverse. Prove every circle point except P lies in the image, and adding P back retains real cardinality.
15. Prove that $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$ if and only if $A \subseteq B$ or $B \subseteq A$.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Appendix 3.

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https://github.com/xinhi6716/computer-science-self-study

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